BERT WEIGHT: Word-Level Sentiment Analysis Through Contextual Token Attribution and Embedding Weighting

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Abstract

This paper introduces BERT WEIGHT, a novel framework that addresses the interpretability limitations of transformer-based sentiment analysis by leveraging word-level contextual information. Unlike conventional approaches that rely solely on the [CLS] token or pooled representations, our methodology assigns sentiment weights to individual tokens, providing fine-grained analysis and enhanced transparency. We propose three distinct methods: (1) an attention-based token attribution approach that utilizes self-attention log-ratios from top layers to identify class-specific token contributions, (2) a word-level average embedding technique that computes sentiment scores across all 12 BERT layers, and (3) a CLS embedding norm-based scoring method that quantifies each term's influence through systematic evaluation of embedding patterns. Extensive experiments on Amazon, IMDB, and Yelp datasets demonstrate that our approach not only outperforms traditional sentiment models but also provides explicit mapping between sentiment predictions and specific words in the text. By moving beyond opaque sentence-level classifications to transparent word-level sentiment attribution, BERT WEIGHT offers a significant advancement for applications where explainability is essential while maintaining competitive performance.

Keywords: *Attention Mechanism, BERT, CLS Embedding, Sentiment Analysis*

Introduction

In recent years, the exponential growth of digital communication has generated vast amounts of textual data on various platforms. Extracting meaningful insights from this unstructured data has become a critical challenge and opportunity across sectors. Among the various Natural Language Processing (NLP) tasks aimed at solving this challenge, sentiment analysis has garnered significant attention due to its practical utility and impact.

The introduction of transformer-based architectures, especially BERT (Bidirectional Encoder Representations from Transformers), has significantly advanced the field. BERT's ability to generate bidirectional contextual word embeddings has resulted in exceptional performance on several NLP benchmarks. However, most current sentiment analysis models use BERT at the sentence or document level, relying on the final [CLS] token representation for classification underutilizing the detailed semantic richness within BERT's internal layers.

Recognizing this gap, we propose a novel methodology titled "BERT WEIGHT"—a sentiment analysis framework that explicitly leverages BERT's word-level contextual embeddings to compute weighted sentiment contributions from individual tokens by assigning them contextual sentiment score. Rest of the paper is structured as section 2 has the literature review, section 3 has proposed methodology, section 4 explains the experimental setup, section 5 shows the results and inferences, and section 6 is conclusion of the paper.

Literature Review

The Contribution of Term Weighting to Text Classification

Historical approaches such as the Bag-of-Words method consider all occurrences of words as equal, whereas methods such as TF-IDF down-weight frequently occurring words emphasizing words that are more distinctive of a document. Nonetheless, statistical methods do not consider context in words.

Sentiment-Specific Problems with Traditional Weighting

To solve this, researchers have modified term weighting for sentiment analysis. K. Ghag and K. Shah(2014) also introduced SentiTFIDF. Martineau and Finin (2009) introduced Delta TF-IDF to emphasize differences between sentiment classes. These models, however, still could not capture dynamic context.

The Rise of Linguistic and Lexicon-Based Models

Language resources such as SentiWordNet, the Opinion Lexicon, and AFINN assisted in improved sentiment classification with the use of previous sentiment ratings to inform term weighting. These lexicon-based

solutions suffered from less coverage, and inadequate flexibility for new data.

The Turn Towards Deep Contextual Representations

Deep learning introduced context into the mix. ELMo employed bidirectional LSTMs to generate context-dependent word embeddings that enhanced sentiment analysis. This was extended by transformer models such as BERT. BERT established new performance benchmarks by leveraging the [CLS] token.

Unresolved Gaps and Limitations

The majority of BERT-based models are very dependent on the [CLS] token, which may mask the contribution of individual words to predictions. This lowers interpretability. In domains where transparency is paramount this is a significant disadvantage. There is an increasing demand for models that leverage the interpretability of old-school term weighting and the deep contextual strength of transformers.

Proposed Methodology

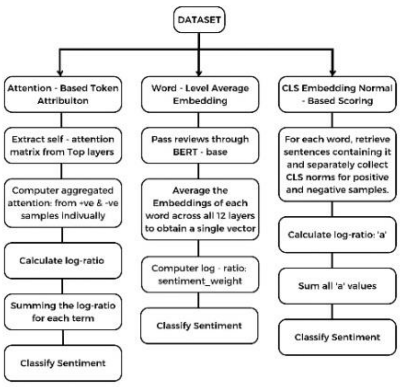

Figure 1: The proposed interpretability framework based on attention and embedding norms

Source: Author

Figure 1 shows the three pipelines and methods we tried out.

Attention-Based Token Attribution

In the first path of our system, we exploit the inherent structure of transformer models of attention scores from the topmost layers by extracting matrix A. This method enables interpretable token-level attribution without many modifications.

Let A_t^i represent the attention weight for term t from the layer i. We define aggregated attention for token t in positively labeled samples as:

$$Attn_t^+ = \frac{1}{|D^+|}\sum_{d\epsilon D^+}\sum_{l\epsilon L_{top}} A_t^{(l)}(d) \quad (1)$$

Where, D^+ is the set of positively tagged reviews and L_{top} represents top layers (typically last 2–4).

Similarly, attention for negatively labeled samples is:

$$Attn_t^- = \frac{1}{|D^-|}\sum_{d\epsilon D^-}\sum_{l\epsilon L_{top}} A_t^{(l)}(d) \quad (2)$$

Where,

D^- is the set of negatively tagged reviews.

Word-level Average Embedding

In the second method, we pass the reviews through a BERT-base-uncased

model and use the embeddings in all the layers to compute the significance of the term.

Let E_t^{avg} be the average of the embeddings of all the 12 layers for a term t. Then:

$$E_t^{avg} = \frac{1}{12}\sum_{l=1}^{12} e_t^l \qquad (3)$$

where e_t^l is the embedding of layer l for term t.

Now, we calculate positive embedding score and negative embedding score for each term.

For positive score:

$$score_t^+ = \frac{1}{R^+}\sum_{r \in R^+} E_t^{avg}(r) \qquad (4)$$

For negative score:

$$score_t^- = \frac{1}{R^-}\sum_{r \in R^-} E_t^{avg}(r) \qquad (5)$$

CLS Embedding Norm-Based Scoring

In the third proposed method, we make use of the [CLS] token embedding score. This method avoids gradient-based explanation techniques and works directly from output embeddings.

S_t^+, S_t^- : Concatenated strings of all positive and negative sentences containing the specific term.

We use the E_{CLS}, the embedding score for the CLS token of each concatenated string using the L2 norm to calculate weight in positive and negative reviews.

For positive strings:

$$pos_weight_t = \left\| E_{CLS}^{S_t^+} \right\| \qquad (6)$$

For negative strings:

$$neg_weight_t = \left\| E_{CLS}^{S_t^-} \right\| \qquad (7)$$

Sentiment Classification

Now for all three methods the last step is finding individual sentiment scores for words.

$$LRA_t = \log_2\left(\frac{Attn_t^+ + \varepsilon}{Attn_t^- + \varepsilon}\right) \qquad (8)$$

$$sentiment_weight_t = log_2\left(\frac{score_t^+ + \delta}{score_t^- + \delta}\right) \qquad (9)$$

$$weight_t = log_2\left(\frac{pos_weight_t + \varepsilon}{neg_weight_t + \varepsilon}\right) \qquad (10)$$

Where ε = 0.001, to avoid divide-by-zero. If the terms score is < 0 it is negative and positive if > 0.

Adding the scores of terms in a review gives us sentiment of that review.

$$Attn_Score = \sum_{j=1}^{n} LRA_{t_j} \qquad (11)$$

$$sentiment_review = \sum_{a=1}^{n} sentiment_weight_{t_a} \qquad (12)$$

$$review_score = \sum_{k=1}^{n} weight_{t_k} \qquad (13)$$

If the total sum comes out < 0 then the review is negative and if > 0, it is positive.

Experimental Setup

We performed experiments on three benchmark datasets: Amazon Reviews, IMDB Movie Reviews, and Yelp Reviews. Each contained a balanced collection of positive and negative

samples with average review lengths from 50 to 300 tokens. Datasets were passed through vanilla BERT in case of the two embedding methods. In the attention attribution method, we used variations of BERT- RoBERTa and DistilBERT (Basbeth, F., and D.H. Fudholi. 2024). We passed all three datasets through the three methods to get a deeper understanding.

Evaluation Protocol

We applied 10-Fold Cross Validation, training on 90% and testing on 10% of each collection in each fold. We compared them with AFINN and VADER to evaluate usefulness.

Performance was evaluated using Accuracy, Precision, Recall, and F1 Score.

Results and Inferences

Figure 2 demonstrates the accuracy of each method and model with the different datasets with the least for RoBERTa and the most for CLS Embedding methodology. Although VADER shows moderately high accuracy, it classifies neutral terms as negative leading to low accuracy.

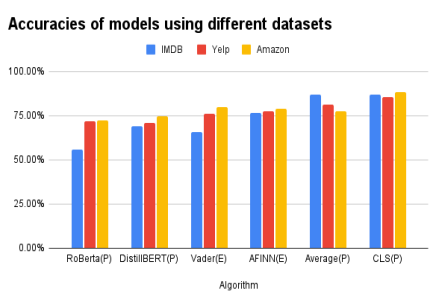

Figure 2: Comparison of Accuracy across all methods and BERT-based models.

Source: Author

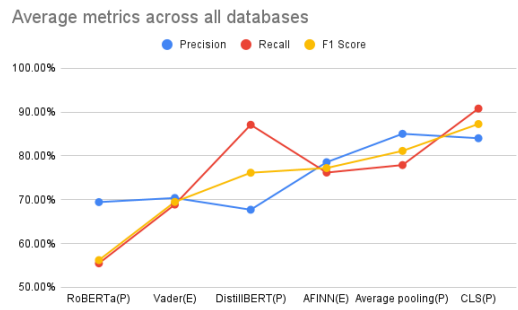

Figure 3: Average metrics of all methods and models across all three databases.

Source: Author

Figure 3 shows us the average precision, recall and F1 score of all the algorithms plotted as a line chart. From that we can see that CLS embedding method has the highest overall metrics compared to others.

In both figures P means proposed and E means existing.

Conclusion

In this paper, we introduced BERT WEIGHT, an interpretable sentiment analysis model that leverages BERT's token-level contextual embeddings via three approaches: attention-based token attribution, word-level average embeddings, and CLS embedding norm scoring. Experimental results on Amazon, IMDB, and Yelp datasets showed that BERT WEIGHT performs much better than conventional sentiment analysis models by identifying fine-grained token importance. Among the approaches, the CLS embedding norm scoring performed the highest overall over accuracy of 86.77%, precision, recall, and F1 score. By uniting the depth of BERT's representations with enhanced explainability, BERT WEIGHT offers a

strong and interpretable option for sentiment analysis task.

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